Quiz 0

Aassume that $y_n = \beta^*{}_0 + x_n \beta^*{}_1 + \varepsilon_n$ for scalar x_n and some fixed $\beta^*{}_0$ and $\beta^*{}_1$. Assume that

- The residuals ε_n are IID with $\mathbb{E}\left[\varepsilon_n\right] = 0$ and $\mathrm{Var}\left(\varepsilon_n\right) = \sigma^2$.
- The regressors x_n are IID with $\mathbb{E}\left[x_n\right] = \mu$ and $\mathrm{Var}\left(x_n\right) = \nu^2$.
- The residuals are independent of the regressors.

You should be able to analyze quantities involving x_n, ε_n, and y_n in terms of the fixed quantities $\beta^*{}_0$, $\beta^*{}_1$, μ, σ, and ν.

- Be able to compute all the conditional variances and expectations you can think of
- Be able to apply the LLN and CLT, including recognizing when the CLT does not converge
- Be able to apply the LLN to random matrices and their inverses, including recognizing when the matrix is or is not invertible.

In addition, you should be able to

- Write out the $\boldsymbol{X}$, $\boldsymbol{Y}$, and $\boldsymbol{\varepsilon}$ matrices for simple linear regressions
- Write out the OLS estimator $\hat{\boldsymbol{\beta}}$ in terms of these quantities
- Be able to use the LLN and CLT to evaluate the limits of
  - $\frac{1}{N}\boldsymbol{X}^\intercal\boldsymbol{X}$
  - $\frac{1}{N}\boldsymbol{X}^\intercal\boldsymbol{Y}$
  - $\frac{1}{\sqrt{N}}\boldsymbol{X}^\intercal\boldsymbol{\varepsilon}$
- Compute $\mathbb{E}\left[\hat{\boldsymbol{\beta}}|\boldsymbol{X}\right]$ and $\mathrm{Cov}\left(\hat{\boldsymbol{\beta}}|\boldsymbol{X}\right)$ in terms of $\mathrm{Cov}\left(\boldsymbol{\varepsilon}\right)$, and plug in for particular values of $\mathrm{Cov}\left(\boldsymbol{\varepsilon}\right)$.

In addititION, you should be able to do all of the above for the special case when there is no x_n and there is only a constant.

Quiz 1

Suppose we observe data points of the form (y_n, s_n, a_n), where a_n is a categorical variable and s_n is either a continous variable or another categorical variable.

You should be able to

- Write the values of the $\boldsymbol{X}$ matrix for a one-hot encoding of a_n
- Write the values of the $\boldsymbol{X}$ matrix for interactions between s_n and a_n
- Understand the meaning of interactions (fitting a different slope for each level)

- Write the regression involving the one-hot encoding with and without a constant term, and undersatnd how the interpretation of the coefficients change

Additionally, for all of these combinations, you should be able to do everything from Quiz 0.

Quiz 2

As a starting point, you should be able to do everything for Quiz 0 with a generic vector $\boldsymbol{x}$ rather than a set of one or two scalars.

You should be familiar with our four assumptions:

- Normality
- Homoskedasticity
- Heteroskedasticity
- Machine learning (IID)

I'll give you what these assumptions are, but you need to be able to understand and apply the definitions.

When we assume that $y_n = \boldsymbol{\beta}^{*\intercal} \boldsymbol{x}_n + \varepsilon_n$, for each of our four assumptions, with enough hints you should be able to do everything from quiz 0, now in terms of $\boldsymbol{\beta}^*$.

For testing, you should be able to analyze the t-statistic for a null of the form ${\boldsymbol{\beta}^*}_k = 0$ for some component k.

- Know what the distribution of the t-statistic is under the null for normality (all you need is to understand its properties, you don't need to memorize anything)
- Know why you would use an F-statistic, and what its distribution is under the normal assumption (all you need is to understand its properties, you don't need to memorize anything)
- Know the distribution of the t-statistic for large N under non-normal assumptions
- Relatedly, understand how to compute the sandwich covariance and how it is different from the standard covariance
- Be able to compute the level of a given test by applying the above results under the null
- Be able to compute the power of a test against a particular alternative by plugging in an alternative $\boldsymbol{\beta}^*$

The question will take the form of undertaking one or more of these tasks under a particular tractable choice of $\boldsymbol{x}$.

Quiz 3

This question will be about a one-dimensional example $y \sim z$ of a machine learning problem, focusing on squared loss.

You should be able to

- Write out the $\boldsymbol{X}$ matrix for a zeroth order spline basis of z
- Be able to plot the basis functions
- Understand intuitively what would change with higher-order splines (no need to know the exact formulas)
- Understand the relationship between the zeroth order spline, partitions of the domain of z, and one-hot encodings
- Know what the fitted value $\hat{y}$ is in each partition and be able to plot them (note: this is a generalization of quiz 1)
- Understand intuitively the costs and benefits of having more or fewer partitions

Additionally, you should be able to do all of Quiz 0 for ridge regression. You should also have an intuitive understanding of the costs and benefits of having a large or small ridge parameter.